

NIHARIKA RATH

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EDUCATION

Master of Astrophysics, Queen Mary University of London September 2025 - 2026

Master's Project: Investigating parametrised post-Newtonian formalism in binary pulsars, focusing on the slow-motion, weak-field limit to test metric theories of gravity and studying their structure and motion in general relativity

Relevant Coursework: AI in Astrophysics and Space Science, Deep Learning

Bachelor of Physics with Astrophysics and Cosmology, King's College London September 2021 - 2024

Undergraduate Project: Analysing M35, M36, and M37 open clusters using photometric methods including photutils, NumPy, and Matplotlib to perform background estimation and aperture photometry and construct their HR diagrams

PROJECTS

Designing and Training a CNN to Classify Features on the Martian Surface

- Developed a CNN-based image classification model for HiRISE Martian surface tiles, addressing class imbalance with class weights and evaluating accuracy, balanced accuracy, macro F1-score, and confusion matrices
- Achieved 94% test accuracy, 88.9% balanced accuracy, and 84.6% macro F1-score

Experimental Investigation of CNN Optimisation for Galaxy Morphology Classification

- Systematically optimised CNNs using TensorFlow/Keras for Galaxy10 dataset morphology classification, experimenting with deeper architectures
- Tested batch normalisation, dropout, L2 regularisation, augmentation, AdamW, and EfficientNetB0 transfer learning to demonstrate their impact on model performance
- Achieved 88% test accuracy, outperforming the baseline model of approximately 74% and transfer learning approaches of 60-67%

Estimating Radiation Belt Diffusion Coefficients using Physics-Informed Neural Networks (PINNs)

- Built two coupled PyTorch neural networks to model phase space density $u(x, t)$ and infer the unknown diffusion coefficient $D(x)$ from Van Allen Probes data using inverse modelling
- Enforced the diffusion PDE through the PINN loss using automatic differentiation, combining data loss, PDE residual loss, regularisation, and positive-output transformations

Forecasting British American Tobacco Stock Volatility Using an LSTM-Based Time-Series Model

- Developed an LSTM model to forecast next-day annualised rolling volatility using historical price-volume data from March 2000 to March 2026 and seven engineered financial features over a 30-day lookback window
- Achieved test-set performance of $RMSE = 0.0195$, $MAE = 0.0126$, and $R^2 = 0.8246$, benchmarked against a persistence baseline

EXPERIENCE

Secretary & Outreach Lead at KCL Space Society, King's College London September 2022 - 2024

- Led STEM outreach initiatives, including school visits, student-facing workshops, and society event coordination
- Designed and delivered an introductory Python workshop for ~20 Year 11 students at Ashcroft Technology Academy interested in pursuing STEM subjects at university

Intern at LIFE - To & Beyond / InnovaSpace UK July 2022 - 2025

- Delivered a World Space Week 2022 talk and panel contribution on "Space Sustainability, Education, Economic Growth, and Inclusion in the Space Sector" and conducted a workshop on scientific communication
- Presented outreach content on Asian lunar exploration, highlighting mission history and scientific contributions across India, China, Japan, and Russia in Lunar Space Week

EXTRA-CURRICULAR ACTIVITIES

- Awarded 1st Dan Black Belt in Shito-Ryu Karate in 2025
- Member of UKSEDS Outreach Team, contributing to space-science workshops and public engagement activities planned for delivery across over 30 schools this summer